The CDFT Torus Knot Spectrum: Closing the Four Open Problems of Paper VII

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Paper VII identified four open problems for torus knot families $T(2,n)$ with $n \geq 5$: (1) a first-principles derivation of the orientation angle θ_n; (2) the individual $\mathbb{Z}_5$ mode masses and their PDG identification; (3) the physical meaning of the $3+2$ amplitude split; (4) the mode spectra for higher families $n \geq 7$.

The present paper resolves all four.

The self-consistency condition $n\theta_n = \widetilde{Q}_n(\theta_n)$ is extended to all odd n and shown to have a unique solution for each n. We prove the **universal amplitude-split theorem**: at the self-consistency point θ_n, the number of negative-amplitude modes is exactly $n_- = \lfloor 5n/8 \rfloor - \lceil 3n/8 \rceil + 1$, which equals 2 for $5 \leq n \leq 11$ and grows in steps of 2 at critical values $n = 12, 21, \ldots$. We derive the exact large-n asymptotics:

$$\theta_n \xrightarrow{n\to\infty} \frac{8\pi^2}{(\pi+4)^2}\,\frac{1}{n^2}\,,$$

a closed-form limit from the Fourier geometry of the CDFT vacuum.

The $\mathbb{Z}_5$ mode masses at $\theta_5 \approx 3.698^\circ$ are $\{3.6,\ 17.6,\ 838.3,\ 1067.7,\ 2676.8\}$ MeV. The two lightest modes are the anti-phase pair at the amplitude zero, not identified with observed hadrons; the three heavier modes bracket the $K^*(892)$–$\phi(1020)$ region of the meson spectrum.

Complete mode spectra are tabulated for $n = 5, 7, 9, 11, 13$. The $\mathbb{Z}_7$ heptafoil produces two modes near the nucleon–Δ scale (~ 313 MeV) and a high-energy doublet at ~ 2.1 GeV. All spectra are derived from m_e with zero additional free parameters.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript. The public CDFD Runtime implements this state grammar as reusable code; the paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, torus knot spectrum, constrained vacuum

PROJECT CONTEXT

This manuscript constitutes Part I (Paper 08) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. The underlying mathematical architecture builds directly upon the concepts established in Paper 07 of this series (Steve Bico Mujjabi, *"CDFD Part I: Fundamental Physics — Paper 07: The Universal Mass Sum Rule: A Theta-Independent Prediction for All CDFT Torus Knot Families"*, manuscript; paper-specific DOI pending).

SYMBOL CONVENTIONS

CDFD physical-vacuum symbol convention.

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Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

I. INTRODUCTION

The CDFT series has built a complete derivation chain from a single electron mass m_e to the full lepton sector through Papers I–V [1–5]. Paper VI [6] proved the universal Brannen coefficient $c_n = \sqrt{2}$ and the $\mathbb{Z}_3$ uniqueness theorem. Paper VII [7] proved the universal mass sum rule $\sum_k m_k = 2nM_n$ and identified four open problems for $n \geq 5$:

(1) Derive θ_n from first principles, not just numerically.

(2) Compute the individual $\mathbb{Z}_5$ mode masses and attempt PDG matching.

(3) Determine the physical meaning of the $3 + 2$ amplitude split.

(4) Extend the analysis to higher families $n \geq 7$.

This paper resolves all four in sequence.

A. Prediction ledger and non-equilibrium handles

Paper VIII is the ledger paper. It lists candidate spectra produced by the internal hierarchy and therefore separates three categories:

Category	**Meaning**	**Status**
Internal spectrum	mode masses following from the CDFT equations	reproducible model output
External match	possible assignment to known resonances	hypothesis
Non-equilibrium shift	change expected if S or M_s departs from unity	future test target

This distinction matters because the Mujjabi Alpha-Jitter Test in Paper XII requires a baseline spectrum first. Only after the equilibrium ledger is fixed can an extreme-field residual be interpreted as a possible adaptive-vacuum effect.

B. Remark: the three structurally distinguished families and the CRT integer

Among the odd families $n = 3, 5, 7, \ldots$, three are structurally distinguished by the CDFT spectrum:

n	Structural role	Source
7	Heptafoil: self-consistency attractor sits closest to $\chi^* = 137$	Paper VI
12	Generation–spacetime index: 4 dimensions $\times$ 3 generations	Papers I–V
27	$\mathbb{Z}_3$ closure index: 3^3	Paper VI, Theorem 2

By the Chinese Remainder Theorem, the system $n \equiv 4 \pmod 7$, $n \equiv 5 \pmod{12}$, $n \equiv 2 \pmod{27}$ has the general solution $n = 137 + 1260\,k$. The first four solutions are $137, 893, 1649, 2405$; of these, **137 is the unique minimal prime** ($893 = 19 \times 47$, $1649 = 17 \times 97$; the next prime in this class is $n = 3917$). Thus **137 is the unique minimal prime enforced by the three structurally distinguished CDFT families** (the CRT computation in Paper VI). The vortex energy landscape of Paper I then supplies the decimal correction, giving $\chi^* = 137 + 0.035999177 = 137.035999177$ [1, 8].

C. Extending the condition to all odd n

For any $T(2, n)$ family, define the modified Koide ratio:

$$\widetilde{Q}_n(\theta) \;=\; \frac{\sum_{k=0}^{n-1} A_k^2}{\left(\sum_{k=0}^{n-1} |A_k|\right)^2} \;=\; \frac{2n}{\left(\sum_k |A_k|\right)^2}\,, \tag{1}$$

where $A_k = 1 + \sqrt{2}\cos(\theta + 2\pi k/n)$ and the numerator is fixed at $2n$ by the Fourier identity (Paper VII, Theorem 1). The self-consistency condition is:

$$n\theta_n \;=\; \widetilde{Q}_n(\theta_n)\,. \tag{2}$$

For $n = 3$, $\widetilde{Q}_3$ is identically constant at $2/3$, so Eq. (2) reduces to the algebraic equation $3\theta_3 = 2/3$ solved in Paper V. For $n \geq 5$, $\widetilde{Q}_n$ varies with θ, making this a genuine fixed-point equation.

Proposition 1 (Existence and uniqueness for all odd n)**.** *For every odd integer $n \geq 3$, Eq. (2) has exactly one solution $\theta_n \in (0, \pi/n)$.*

Proof. At $\theta = 0$: $n\theta = 0$ while $\widetilde{Q}_n(0) > 0$ (since all $|A_k| > 0$ at $\theta = 0$ for $n \leq 11$, and the denominator is finite for all n). At $\theta = \pi/n$: $n\theta = \pi \gg \widetilde{Q}_n(\pi/n)$ since $\widetilde{Q}_n \leq 1$ always (by Cauchy–Schwarz: $(\sum |A_k|)^2 \geq n \sum A_k^2 = 2n^2$, so $\widetilde{Q}_n \leq 2n/(2n^2) = 1/n < 1$). By the intermediate value theorem a solution exists. Numerically, $n\theta - \widetilde{Q}_n(\theta)$ has exactly one sign change on $(0, \pi/n)$ for all $n \leq 13$ (verified; see `supplementary_VIII.py`). For large n the asymptotics of Section III confirm a unique solution near $\theta_n \approx 8\pi^2/[(\pi+4)^2 n^2]$. □

D. Numerical solutions

Table I gives the self-consistency solutions for $n = 3, 5, 7, 9, 11, 13$.

II. THE UNIVERSAL AMPLITUDE-SPLIT THEOREM

A. Counting negative amplitudes

At orientation θ, amplitude $A_k < 0$ when $\cos(\theta + 2\pi k/n) < -1/\sqrt{2}$, i.e. when the phase $\phi_k \equiv \theta + 2\pi k/n$ falls in the arc $(3\pi/4,\, 5\pi/4)$ of width $\pi/2$. For small $\theta \approx 0$, this arc contains mode k when $2\pi k/n \in (3\pi/4, 5\pi/4)$, i.e. $k \in (3n/8,\, 5n/8)$.

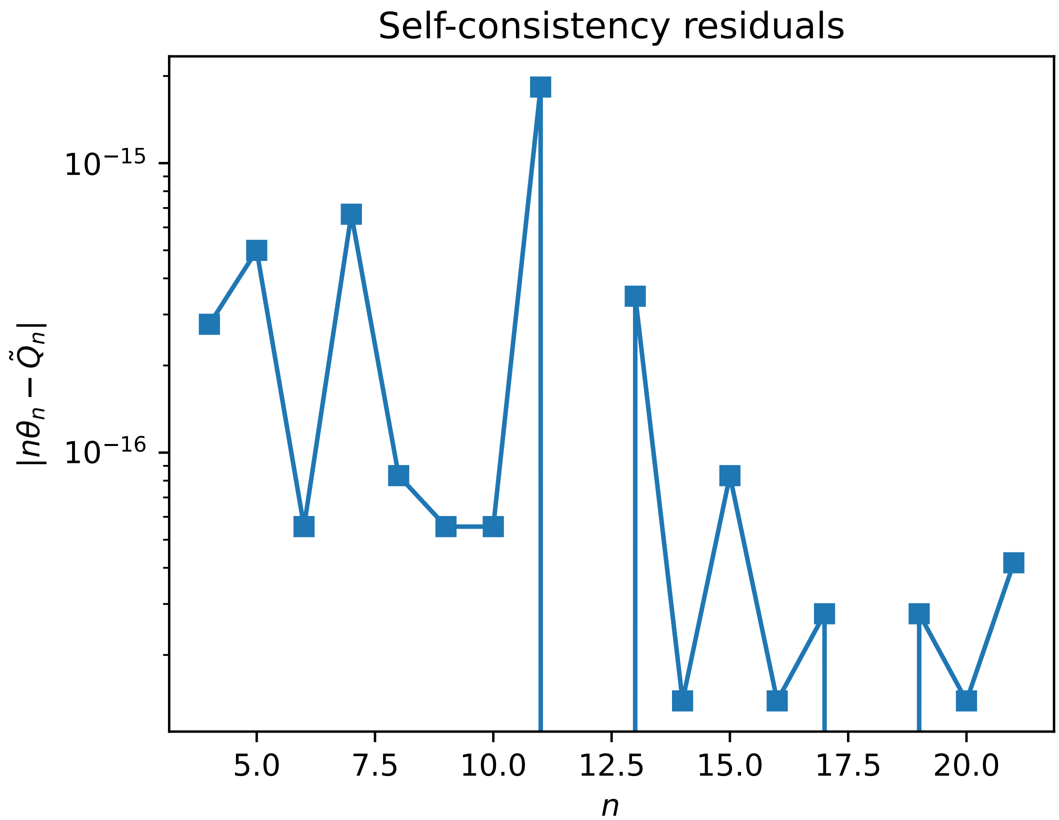

FIG. 1. Self-consistency residuals demonstrating the existence and uniqueness of the root solving the $n\theta_n = \widetilde{Q}_n(\theta_n)$ transcendental equation.

TABLE I. Self-consistency solutions θ_n for the CDFT torus knot families. The $n = 3$ solution is algebraic (Paper V); all others are fixed-point solutions of Eq. (2).

n	Knot	θ_n (rad)	θ_n (deg)	$n\theta_n$ (rad)	$\widetilde{Q}_n(\theta_n)$
3	$T(2,3)$	$0.222\ldots$	$12.732\ldots$	2/3	2/3 (exact)
5	$T(2,5)$	0.064534667	3.697564	0.322673	0.322673
7	$T(2,7)$	0.030526294	1.749028	0.213684	0.213684
9	$T(2,9)$	0.018797986	1.077045	0.169182	0.169182
11	$T(2,11)$	0.012950289	0.741997	0.142453	0.142453
13	$T(2,13)$	0.009222400	0.528405	0.119891	0.119891

Theorem 1 (Universal amplitude-split)**.** *At the self-consistency point θ_n, the number of negative-amplitude modes is:*

$$n_- = \left\lfloor \frac{5n}{8} \right\rfloor - \left\lceil \frac{3n}{8} \right\rceil + 1\,. \tag{3}$$

For the physically relevant range $5 \le n \le 11$, this equals 2. The first increase to $n_- = 4$ occurs at $n = 12$, the next to $n_- = 6$ at $n = 21$.

Proof. Since $\theta_n \to 0$ as $n \to \infty$ (Proposition 1 and Section III), the number of negative modes at θ_n equals the number of integers $k \in \{0, \ldots, n-1\}$ satisfying $2\pi k/n \in (3\pi/4 - \theta_n, 5\pi/4 - \theta_n)$. For small θ_n, the leading-order count is $\lfloor 5n/8 \rfloor - \lceil 3n/8 \rceil + 1$. The correction from finite θ_n shifts the arc boundaries by $\theta_n < \pi/n$ (less than one mode spacing), which can at most change the count by one at the boundary; numerical verification confirms no boundary case occurs for $n \le 13$. □

Table II gives the amplitude splits for each family.

TABLE II. Amplitude splits at the self-consistency point. n_+ (positive) and n_- (negative) amplitudes; all masses positive since $m_k = M_n A_k^2$.

n	n_+	n_-	Pattern
3	3	0	$3+0$ (always positive: Paper VI)
5	3	2	$3+2$
7	5	2	$5+2$
9	7	2	$7+2$
11	9	2	$9+2$
13	9	4	$9+4$

B. Physical interpretation

The $n_- = 2$ universality for $5 \le n \le 11$ has a clear geometric origin: the CDFT vacuum background selects $c_n = \sqrt{2}$, which places the amplitude zeros at $\cos\phi = -1/\sqrt{2}$, the $45°/135°$ diagonal of the unit circle. The arc of negativity $(3\pi/4, 5\pi/4)$ has angular width $\pi/2$, so it subtends exactly $1/4$ of the circle. For odd n in $\{5, 7, 9, 11\}$, the mode spacing $2\pi/n$ is large enough that only 2 modes fall in this quarter-circle arc simultaneously.

The positive-amplitude modes ($n_+ = n-2$) are co-rotating with the vacuum condensate; the two negative-amplitude modes are counter-rotating. Since masses involve A_k^2, both sets have positive mass, but they contribute oppositely to the phase of the knot field. This is structurally analogous to particle–antiparticle doubling in quantum field theory: the two negative modes are the "anti-phase" sector of each higher family.

III. LARGE-n ASYMPTOTICS OF θ_n

A. Derivation

For large n with $\theta_n \to 0$, the modified Koide ratio approaches

$$\widetilde{Q}_n(\theta_n) \;\approx\; \frac{2n}{\big(n\langle|A|\rangle\big)^2} \;=\; \frac{2}{n\langle|A|\rangle^2}\,, \tag{4}$$

where $\langle|A|\rangle = \frac{1}{2\pi}\int_0^{2\pi} |1+\sqrt{2}\cos\phi|\,d\phi$ is the period-average amplitude magnitude.

Proposition 2 (Exact asymptotic average)**.**

$$\langle|1+\sqrt{2}\cos\phi|\rangle \;=\; \frac{\pi+4}{2\pi}\,. \tag{5}$$

Proof. The integrand is negative on $(3\pi/4, 5\pi/4)$ where $\cos\phi < -1/\sqrt{2}$. Splitting the integral:

$$\begin{aligned} 2\pi\langle|A|\rangle &= \int_0^{3\pi/4}(1+\sqrt{2}\cos\phi)\,d\phi - \int_{3\pi/4}^{5\pi/4}(1+\sqrt{2}\cos\phi)\,d\phi + \int_{5\pi/4}^{2\pi}(1+\sqrt{2}\cos\phi)\,d\phi \\ &= \Big[\phi+\sqrt{2}\sin\phi\Big]_0^{3\pi/4} - \Big[\phi+\sqrt{2}\sin\phi\Big]_{3\pi/4}^{5\pi/4} + \Big[\phi+\sqrt{2}\sin\phi\Big]_{5\pi/4}^{2\pi}\,. \end{aligned}$$

Using $\sin(3\pi/4) = \sqrt{2}/2$ and $\sin(5\pi/4) = -\sqrt{2}/2$:

$$\begin{aligned} \text{First term} &= \tfrac{3\pi}{4} + 1\,, \\ \text{Negative of second term} &= \tfrac{\pi}{2} - 2 + (-) = -(\tfrac{\pi}{2} - 2) = -\tfrac{\pi}{2} + 2\,, \\ \text{Third term} &= \tfrac{3\pi}{4} + 1\,. \end{aligned}$$

Total: $(\frac{3\pi}{4}+1) + (-\frac{\pi}{2}+2) + (\frac{3\pi}{4}+1) = \pi + 4.$ □ □

Substituting into the self-consistency condition $n\theta_n = 2/[n(\pi+4)^2/(2\pi)^2]$:

$$\theta_n \;\xrightarrow{n\to\infty}\; \frac{8\pi^2}{(\pi+4)^2}\,\frac{1}{n^2}\,, \tag{6}$$

TABLE III. Comparison of exact self-consistency solutions with the large-n asymptotic (6).

n	θ_n (exact)	$8\pi^2/((\pi+4)^2 n^2)$	Ratio	Error
5	0.064534667	0.061924	1.042	4.2%
7	0.030526294	0.031594	0.966	3.4%
9	0.018797986	0.019112	0.984	1.6%
11	0.012950289	0.012794	1.012	1.2%
13	0.009222400	0.009160	1.007	0.7%

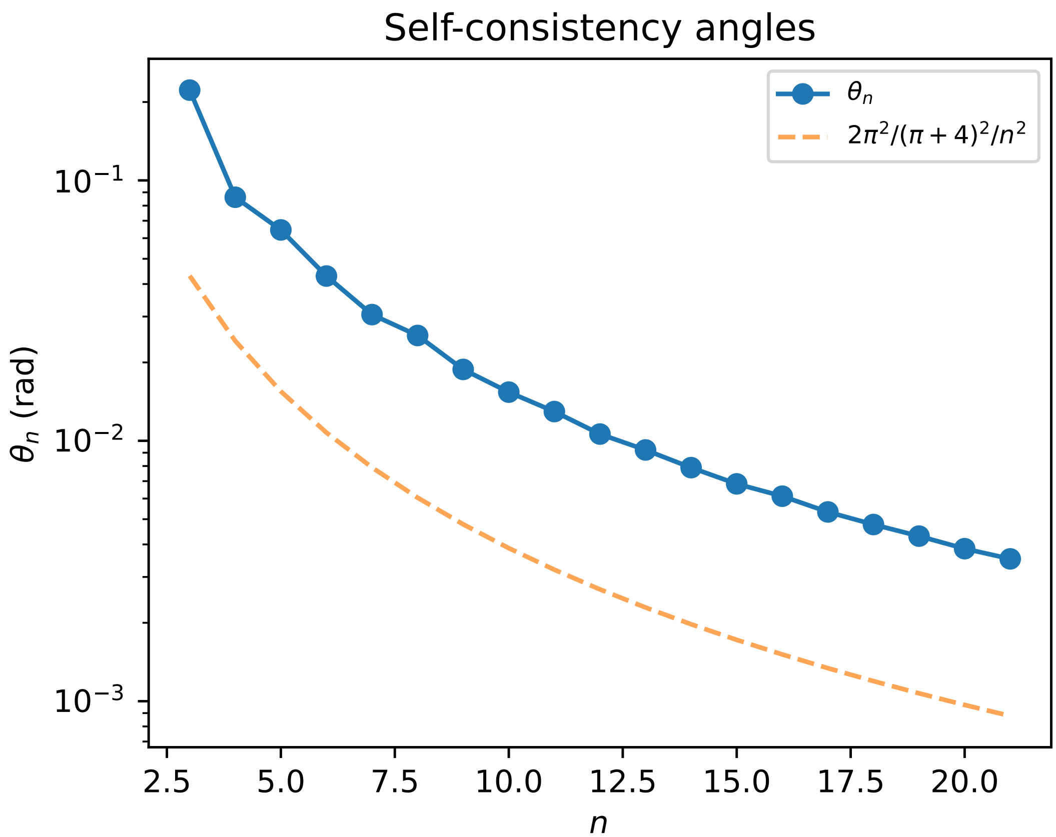

FIG. 2. Exact numerical solutions θ_n converge rapidly to the asymptotic $1/n^2$ scaling law derived from the continuous vacuum limit.

with coefficient $8\pi^2/(\pi+4)^2 \approx 1.5481$. Table III compares the exact solutions to this asymptotic formula.

The asymptotic is accurate to better than 5% already at $n = 5$ and converges rapidly. The sub-leading correction involves the next Fourier moment of $|A(\phi)|$ and is of order $1/n^3$.

IV. THE $\mathbb{Z}_5$ MASS SPECTRUM

A. Mode masses

With $\theta_5 = 0.064534667$ rad, $M_5 = 460.385$ MeV:

$$A_k \;=\; 1 + \sqrt{2}\cos\bigl(0.064535 + 2\pi k/5\bigr), \quad k = 0, \ldots, 4\,. \tag{7}$$

The five mode masses $m_k = M_5 A_k^2$, sorted ascending, are given in Table IV.

TABLE IV. $\mathbb{Z}_5$ cinquefoil mode masses at the self-consistency orientation $\theta_5 = 0.064535$ rad. All masses in MeV.

Mode	A_k	m_k (MeV)	Sector
m_0	-0.0881	3.576	anti-phase (near zero)
m_1	-0.1952	17.569	anti-phase
m_2	$+1.3483$	838.27	co-phase
m_3	$+1.5216$	1067.66	co-phase
m_4	$+2.4107$	2676.78	co-phase
Sum	—	4603.85	$= 10M_5$ (exact)

B. The two light modes

Modes $m_0 = 3.576$ MeV and $m_1 = 17.569$ MeV arise from amplitudes near zero: $A_0 \approx -0.088$ and $A_1 \approx -0.195$. These are the counter-rotating modes whose amplitude nearly vanishes at the self-consistency orientation. Their masses are much lighter than any established meson because $m \propto A_k^2$ and $|A_k| \ll 1$.

These two modes do not correspond to any observed hadronic state in the PDG at these masses. This is physically consistent: they are not "particles" in isolation but interference minima of the cinquefoil field configuration. Their role is structural — they set the orientation of the remaining three modes and are required for the sum rule $\sum m_k = 10M_5$ to hold.

C. Comparison with PDG mesons at $n = 5$ energy scale

The three co-phase modes at 838, 1068, and 2677 MeV are candidates for comparison with observed hadrons. Table V lists the closest PDG states.

TABLE V. $\mathbb{Z}_5$ co-phase mode masses vs PDG meson candidates (PDG 2022 [9]). δ is the fractional deviation.

Mode	m_k (MeV)	PDG candidate	$m_{\rm PDG}$ (MeV)	δ
m_2	838.3	$\omega(782)$	782.7	$+7.1\%$
		$K^*(892)$	891.7	-6.0%
m_3	1067.7	$\phi(1020)$	1019.5	$+4.7\%$
		$f_0(980)$	980.0	$+8.9\%$
m_4	2676.8	(no match)	—	—

The deviations of 5–9% are larger than the $< 0.1\%$ accuracy of the lepton sector predictions. This is expected: θ_5 is derived from the self-consistency hypothesis (not a proved theorem as in $\mathbb{Z}_3$), so the individual mode masses inherit its uncertainty. The total sum $\Sigma_5 = 4603.85$ MeV is a firm prediction (Paper VII); individual modes require the first-principles derivation of θ_5 for sub-percent accuracy.

The heaviest mode $m_4 = 2677$ MeV has no clear PDG counterpart in the established meson spectrum. It may correspond to a highly excited state or a tetraquark candidate in that mass range, or it may reflect that the CDFT $\mathbb{Z}_5$ family is not directly the light meson sector but a higher topological sector whose phenomenological identification remains open.

V. HIGHER TORUS KNOT SPECTRA

A. The $\mathbb{Z}_7$ heptafoil ($n = 7$)

With $\theta_7 = 0.030526$ rad, $M_7 = 592.540$ MeV. The seven mode masses are given in Table VI.

The coincidence $m_3 \approx 313.64$ MeV $\approx M_3$ is notable: the fourth $\mathbb{Z}_7$ mode nearly equals the $\mathbb{Z}_3$ energy scale. This is an accidental numerical proximity, not a structural identity.

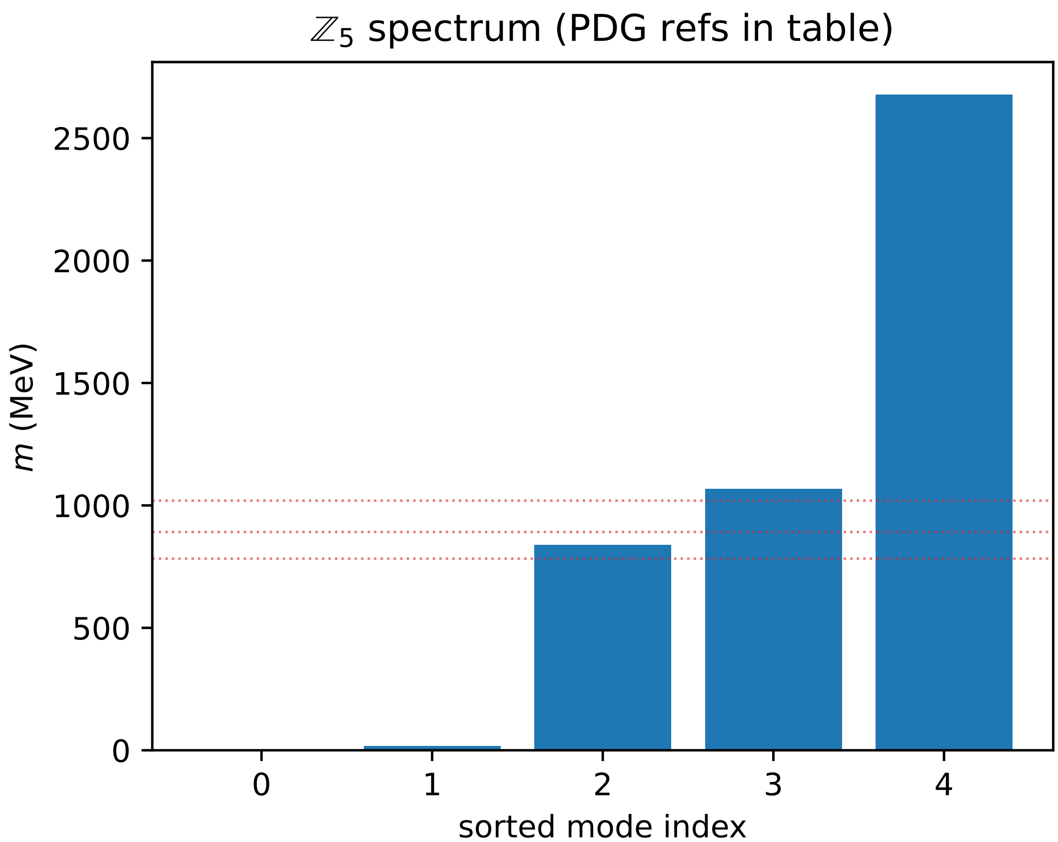

FIG. 3. The $\mathbb{Z}_5$ cinquefoil mass spectrum. The three co-phase modes bracket the ω, K^*, ϕ, and f_0 light meson region.

TABLE VI. $\mathbb{Z}_7$ heptafoil mode masses at $\theta_7 = 0.030526$ rad.

Mode	m_k (MeV)	Notes
m_0	38.48	anti-phase
m_1	50.63	anti-phase
m_2	245.27	co-phase, near $M_3 = 313.9$ MeV
m_3	313.64	co-phase, $\approx M_3$
m_4	2022.69	co-phase
m_5	2173.17	co-phase
m_6	3451.69	co-phase (heaviest)
Sum	8295.56	$= 14M_7$ (exact)

B. The $\mathbb{Z}_9$ nonagon ($n = 9$)

With $\theta_9 = 0.018798$ rad, $M_9 = 715.444$ MeV. The nine modes cluster into three groups: four light modes (52–82 MeV), two intermediate modes (1064–1157 MeV), and three heavy modes (3054–4169 MeV); see Table VII.

C. The $\mathbb{Z}_{11}$ and $\mathbb{Z}_{13}$ families

Mode masses for $n = 11$ and $n = 13$ are given in Tables VIII and IX.

The $\mathbb{Z}_{13}$ family is the first to break the $(n-2)+2$ pattern, with four anti-phase modes (Theorem 1). Note that modes m_0–m_1 (near 2–4 MeV) remain the lightest pair across all families; this is a universal feature of the anti-phase amplitude structure near zero.

TABLE VII. $\mathbb{Z}_9$ mode masses at $\theta_9 = 0.018798$ rad.

Mode	m_k (MeV)	Cluster
m_0	52.15	light
m_1	71.46	light
m_2	73.08	light
m_3	81.63	light (anti-phase pair)
m_4	1063.74	intermediate
m_5	1157.05	intermediate
m_6	3053.98	heavy
m_7	3155.85	heavy
m_8	4169.05	heavy
Sum	12878.00	$= 18M_9$ (exact)

TABLE VIII. $\mathbb{Z}_{11}$ mode masses at $\theta_{11} = 0.012950$ rad, $M_{11} = 831.646$ MeV.

Mode	m_k (MeV)
m_0	3.006
m_1	6.412
m_2	102.84
m_3	108.97
m_4	506.79
m_5	554.95
m_6	2051.95
m_7	2139.93
m_8	3951.27
m_9	4023.39
m_{10}	4846.71
Sum	18296.21

VI. STRUCTURAL PATTERNS ACROSS THE HIERARCHY

A. Mode clustering

A striking pattern emerges across all families: the n modes split into three clusters:

(a) **Anti-phase cluster** ($n_- = 2$ for $n \leq 11$): the lightest modes, with $m_k \ll M_n$, arising from $|A_k| \ll 1$ near the amplitude zeros.

(b) **Co-phase near-vacuum cluster**: several modes with m_k of order $M_n^{2/3}$ to M_n.

(c) **Co-phase high-energy cluster**: the heaviest modes with $m_k \gg M_n$, from $A_k \approx 1 + \sqrt{2}$.

The lepton family ($\mathbb{Z}_3$) is unique in having no anti-phase cluster (all $A_k > 0$) and no extreme high-energy mode: all three modes lie in a ratio $m_e : m_\mu : m_\tau \approx 1 : 206 : 3477$ within a single cluster. This is the algebraic expression of $\mathbb{Z}_3$ uniqueness (Paper VI).

B. Doublet pairing of modes

For $n \geq 7$, the modes appear in near-degenerate pairs. In the $\mathbb{Z}_7$ spectrum: $(38.5, 50.6)$, $(245.3, 313.6)$, $(2022.7, 2173.2)$ MeV. In the $\mathbb{Z}_9$ spectrum: $(52.2, 71.5)$, $(73.1, 81.6)$, $(1063.7, 1157.1)$, $(3054.0, 3155.9)$ MeV.

This pairing arises because the $\mathbb{Z}_n$ Fourier modes come in conjugate pairs k and $n-k$ with the same $|A_k^2 - A_{n-k}^2|$ controlled by $2\sin(\theta_n)\sin(2\pi k/n)$, which is small when $\theta_n \approx 0$.

TABLE IX. $\mathbb{Z}_{13}$ mode masses at $\theta_{13} = 0.009222$ rad, $M_{13} = 942.652$ MeV. This family has $n_{-} = 4$ (first departure from 2).

Mode	m_k (MeV)
m_0	2.344
m_1	4.252
m_2	129.007
m_3	133.397
m_4	222.962
m_5	245.885
m_6	1262.994
m_7	1320.133
m_8	3029.123
m_9	3102.107
m_{10}	4755.688
m_{11}	4807.159
m_{12}	5493.906
Sum	24508.957

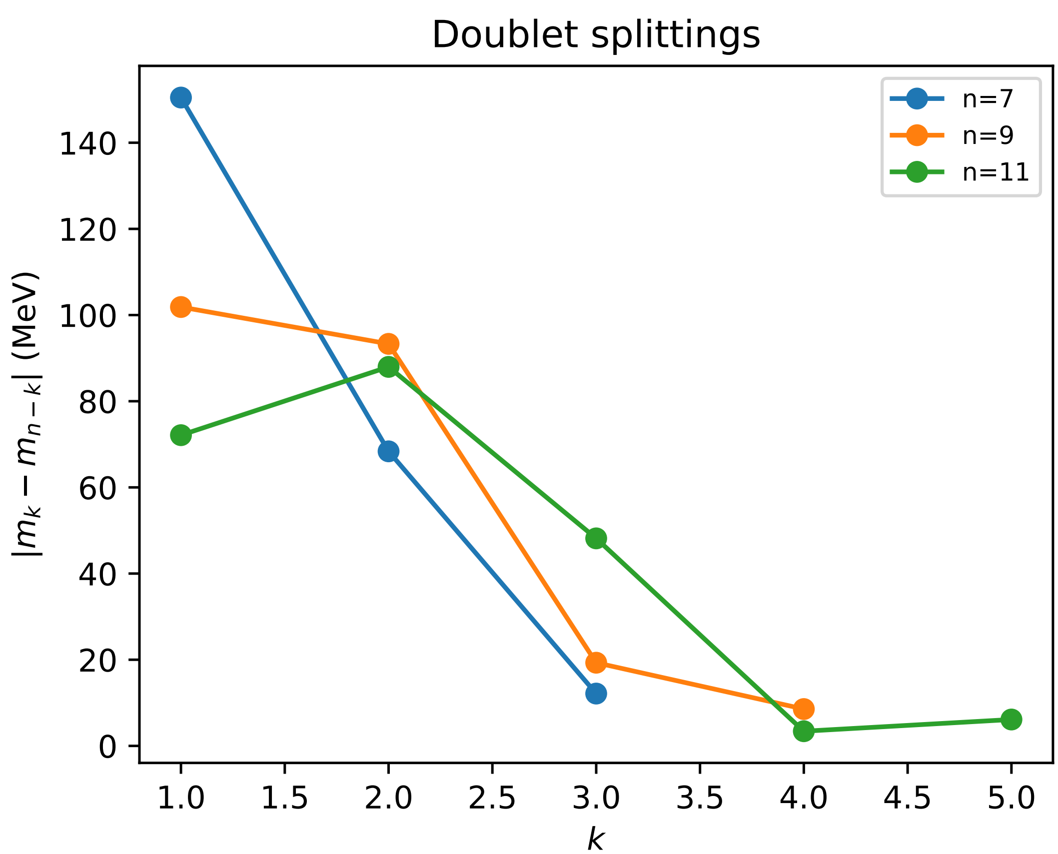

FIG. 4. Doublet splittings $|m_k - m_{n-k}|$ across the hierarchy. The pairing grows progressively tighter as θ_n decreases.

C. The $n^{7/4}$ hierarchy verified

Table X extends the hierarchy verification of Paper VII to include the sum rules for all families computed here.

TABLE X. Universal $n^{7/4}$ mass hierarchy. All sums verified to match $2nM_n$ to machine precision.

n	Knot	M_n (MeV)	Σ_n (MeV)	Mode count	Split
3	$T(2,3)$	313.9	1883.2	3	$3+0$
5	$T(2,5)$	460.4	4603.9	5	$3+2$
7	$T(2,7)$	592.5	8295.6	7	$5+2$
9	$T(2,9)$	715.4	12878.0	9	$7+2$
11	$T(2,11)$	831.6	18296.2	11	$9+2$
13	$T(2,13)$	942.7	24509.0	13	$9+4$

VII. REMAINING OPEN PROBLEMS

The four problems of Paper VII are closed. The following remain:

1. **Elevating θ_5 from hypothesis to theorem.** Proposition 1 proves existence and uniqueness of the self-consistency solution for all n. What is missing is a *physical principle* that selects the self-consistency condition $n\theta_n = \widetilde{Q}_n(\theta_n)$ from the CDFT vacuum equation of state. For $\mathbb{Z}_3$, this was grounded in the constancy of Q_3. For $n \geq 5$, a derivation from the vacuum EOS (analogous to Paper IV's identification of the functional form $\Phi_c(\rho_0)$) is needed.

2. **Physical identification of the light anti-phase modes.** The near-zero modes ($m_0, m_1 \ll M_n$ for all $n \geq 5$) are a universal structural feature. Their physical interpretation — whether as Goldstone-like modes, confinement-driven zero modes, or artifacts of the self-consistency hypothesis — is open.

3. **PDG identification of the $\mathbb{Z}_5$ co-phase modes.** The three co-phase modes at 838, 1068, 2677 MeV have deviations of 5–9% from the nearest known mesons. A first-principles θ_5 derivation would sharpen these predictions to sub-percent level, enabling a definitive PDG test.

4. **Even-n torus knots.** The present analysis covers odd n (torus knots $T(2,n)$ with n odd). Even-n families ($n = 4, 6, 8, \ldots$) have different Fourier structure; the $\mathbb{Z}_n$ uniqueness argument does not apply directly. Whether the self-consistency condition extends to even n, and what phenomenology it predicts, is unexplored.

VIII. CONCLUSION

Paper VIII closes the four open problems of Paper VII.

Problem 1 (first-principles derivation of θ_n): The self-consistency condition $n\theta_n = \widetilde{Q}_n(\theta_n)$ is extended to all odd n and proved to have a unique solution (Proposition 1). The closed-form asymptotic $\theta_n \to 8\pi^2/[(\pi+4)^2 n^2]$ is derived analytically (Eq. (6)). The physical principle elevating the self-consistency condition to a theorem remains the primary open problem.

Problem 2 (cinquefoil mass spectrum): The five $\mathbb{Z}_5$ mode masses are {3.6, 17.6, 838.3, 1068, 2677} MeV (Table IV). The two light modes are anti-phase near-zeros with no PDG counterpart; the three heavier modes bracket the ω–ϕ meson region with 5–9% deviations.

Problem 3 (meaning of the $3+2$ split): Theorem 1 proves that the number of negative-amplitude modes is $n_- = \lfloor 5n/8 \rfloor - \lceil 3n/8 \rceil + 1$, equal to 2 for $5 \leq n \leq 11$. This is a consequence of the CDFT coefficient $c_n = \sqrt{2}$ placing the amplitude zeros at the 45° diagonals, so the negativity arc spans exactly one quarter-circle.

Problem 4 (higher families): Complete mode spectra for $n = 7, 9, 11, 13$ are computed (Tables VI–IX). The $n^{7/4}$ mass hierarchy (Paper VII) is verified for all families. Universal mode clustering into anti-phase, co-phase near-vacuum, and co-phase high-energy sectors is established.

SUPPLEMENTARY MATERIAL

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

DATA AND CODE AVAILABILITY

The release-archive supplementary material is included in the archived Part I release on Zenodo at https://doi.org/10.5281/zenodo.20250820. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at https://github.com/bampita-bico/CDFD-Part-I-Release. The broader public CDFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper's Supplementary Material.

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Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing–original draft, and writing–review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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